

Validating the Higgs Boson

Overview

In a group of four, I worked on a simulated collider-data analysis designed to test whether a Higgs-like signal was supported by the data.

Method

We generated and analysed collision data using NumPy and Matplotlib. The fitting workflow used maximum-likelihood estimation and χ^2 minimisation to compare signal-and-background models.

Outcome

The reduced χ^2 improved from 2.83 to 1.13 after including the Higgs contribution at the simulated mass. The analysis was documented in a formal group report; this public page and its PDF are a concise project summary, not the original report.

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